

# COMPREHENSIVE OVERVIEW OF TRANSACTION CARDS AND BANK ACCOUNTS

- Presented by

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# What are Transaction Cards?



- **Transaction cards are financial instruments that enable cashless payments and electronic fund transfers.**
- **Used for secure and convenient payments across various platforms (online, in-store, ATMs).**

# History and Evolution of Transaction Cards

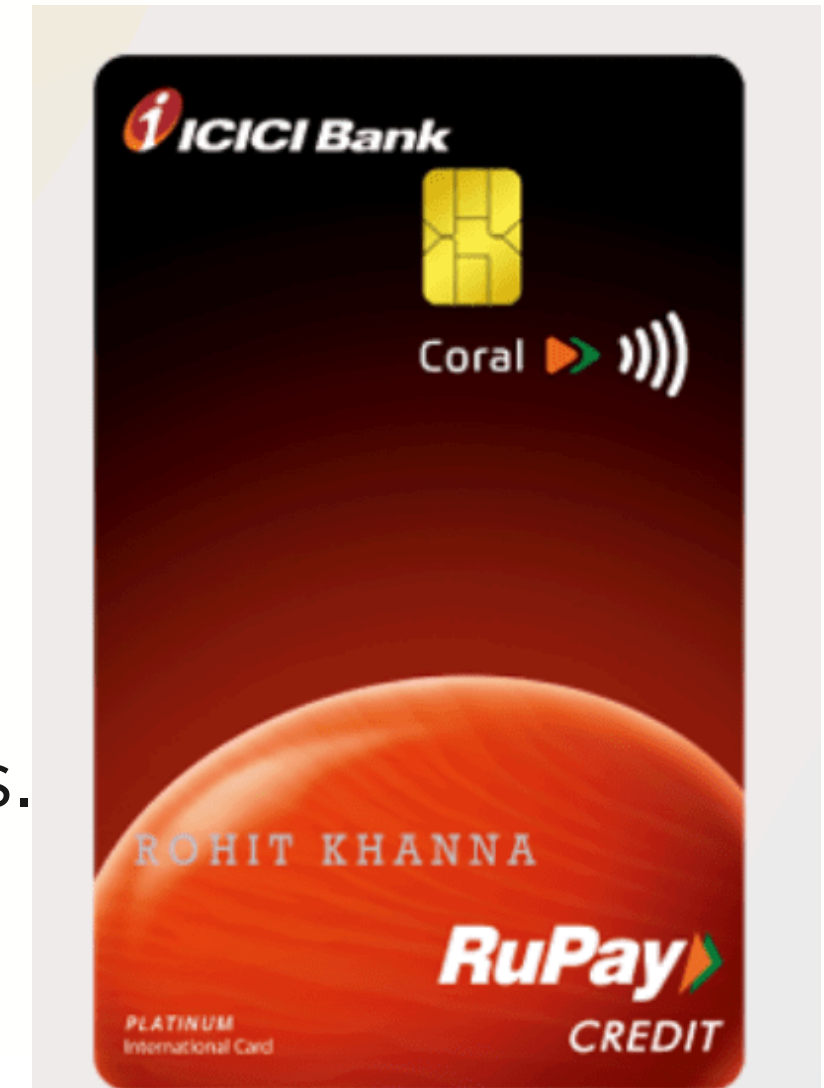


|                                    |       |                                                                 |
|------------------------------------|-------|-----------------------------------------------------------------|
| <b>First charge card</b>           | 1920s | Metal coins and cards                                           |
| <b>Cheque guarantee card</b>       | 1943  | Abbreviated portable letter of credit                           |
| <b>First universal credit card</b> | 1950  | Introduced by <b>Diners Club</b>                                |
| <b>ATM Cards</b>                   | 1967  | Issued by <b>Barclays</b> in <b>London</b>                      |
| <b>Magnetic Strip Cards</b>        | 1972  | issued by <b>Lloyds Bank</b>                                    |
| <b>Smart Card Technology</b>       | 1970s | Was adapted for use as a payment card                           |
| <b>EMV Chip</b>                    | 1990s | Europay, Mastercard, and Visa (EMV) introduced chip-based cards |
| <b>Contactless Technology</b>      | 2000s | NFC-based payments for faster transactions.                     |

# Types of Cards

## Credit Cards

- Linked to a line of credit provided by the issuer, allowing the cardholder to borrow for purchases or cash advances.
  - Interest charged on unpaid balance after due date.
  - Builds credit history and improves credit score if managed well.
- Used For,
  - Making large purchases.
  - Emergency expenses.
  - Earning rewards and cashback.



# Types of Cards

## Debit Cards

- Linked to the cardholder's bank account; funds are withdrawn immediately upon making a purchase.
  - Direct access to bank funds.
- - Works as an ATM card and cheque guarantee card.
  - No interest charged.
- Used For,
  - Everyday purchases.
  - ATM withdrawals.
  - Direct bank-to-merchant payments.





# Types of Cards

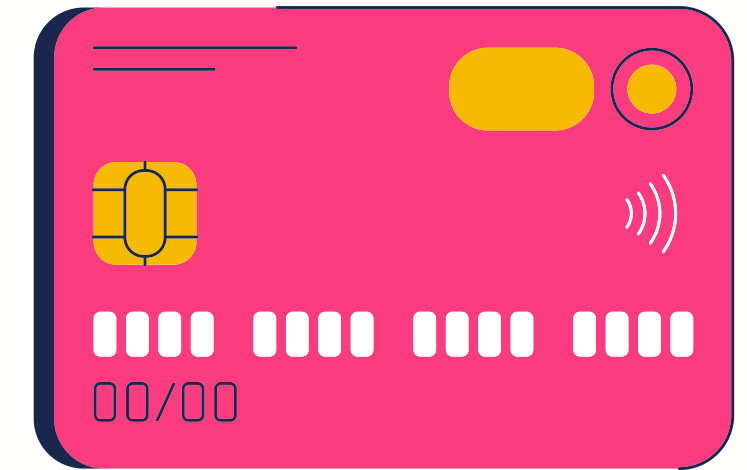
## Prepaid Cards

- Pre-loaded with a fixed amount of money; not linked to a bank account or credit line.
  - Funds stored directly on the card or issuer's database.
  - Can be reloaded in some cases.
  - No overdraft facility.
- Used For,
  - Gift cards.
  - Travel cards.
  - Expense control for minors or employees.





# Types of Cards

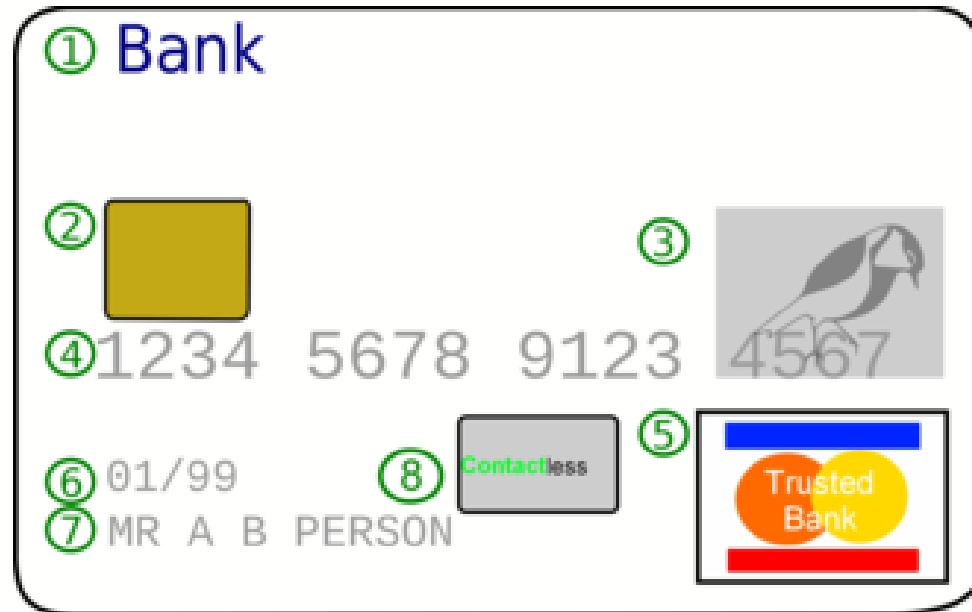


## Virtual Credit Cards

- Digital credit cards that generate unique 16-digit card numbers and expiration dates for each transaction.
- - Enhanced security and fraud protection.
  - Single-use or limited-use card numbers.
  - Easier expense management.
- Used For,
  - Online purchases.
  - Business transactions.
  - Secure vendor payments.

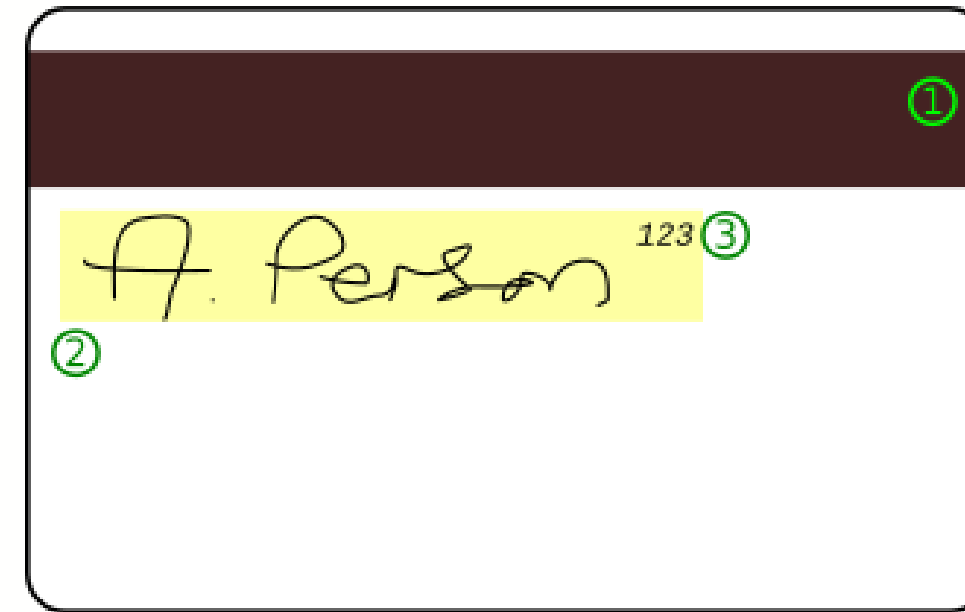


# Elements in Transaction Cards



● An example of the front of a typical debit card:

1. Issuing bank logo
2. EMV chip
3. Hologram
4. Card number (PAN)
5. Card brand logo
6. Expiration date
7. Cardholder's name



● An example of the reverse side of a typical debit card:

1. Magnetic stripe
2. Signature strip panel
3. Card Security Code



# Elements in Transaction Cards

## Card Number

IINs (Issuer Identification Number) and PANs have a certain level of internal structure and share a common numbering scheme set by ISO/IEC 7812.

- **Structure:**

- IIN: First 6 or 8 digits (first digit is Major Industry Identifier - MII).
- Account Identifier: Up to 12 digits (varies).
- Check Digit: Single digit (Luhn algorithm).

**Updates:** IIN length extended to 10 digits (ISO/IEC 7812, 2017); PAN remains 10–19 digits.

- **Card Number Length:**

- American Express: 15
- RuPay: 16
- Mastercard: 16
- Visa: 13/16/19



Partial IIN on a credit card  
(both printed and embossed)

# Elements in Transaction Cards

## Expiry Date:

- **Format:** (MM/YY or MM-YY)



# Elements in Transaction Cards

- CVV Code

- **Card Verification Value** is a series of numbers that, in addition to the bank card number, is printed on a credit or debit card.
- The CVV is used as a security feature for card not present transactions



The CVV is located on the back of Mastercard, Visa, Discover, Diners Club, and JCB credit or debit cards



On American Express cards, the CVV is a printed, group of four digits on the front towards the right

# Elements in Transaction Cards

- **CVV Naming**

| Code                           | Card Type                     | Position                                                | Description                               |
|--------------------------------|-------------------------------|---------------------------------------------------------|-------------------------------------------|
| CSC (Card Security Code)       | Debit Cards, American Express | Back (3 digits)                                         | Also called 3CSC                          |
| CVC (Card Validation Code)     | Mastercard                    | Back (3 digits)                                         | -                                         |
| CVV (Card Verification Value)  | Visa                          | Back (3 digits)                                         | -                                         |
| CID (Card Identification Code) | Discover, American Express    | Front (4 digits) for AmEx, Back (3 digits) for Discover | Also called Card ID/Identification Number |
| CVD (Card Verification Data)   | Discover                      | Back (3 digits)                                         | -                                         |
| CVN (Card Validation Number)   | China UnionPay, Google Ads    | Back (3 digits)                                         | Also called Card Verification Number      |
| CCV (Card Code Verification)   | Canada                        | Back (3 digits)                                         | -                                         |

# Elements in Transaction Cards

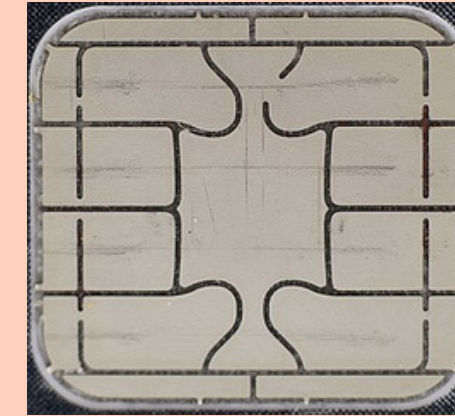
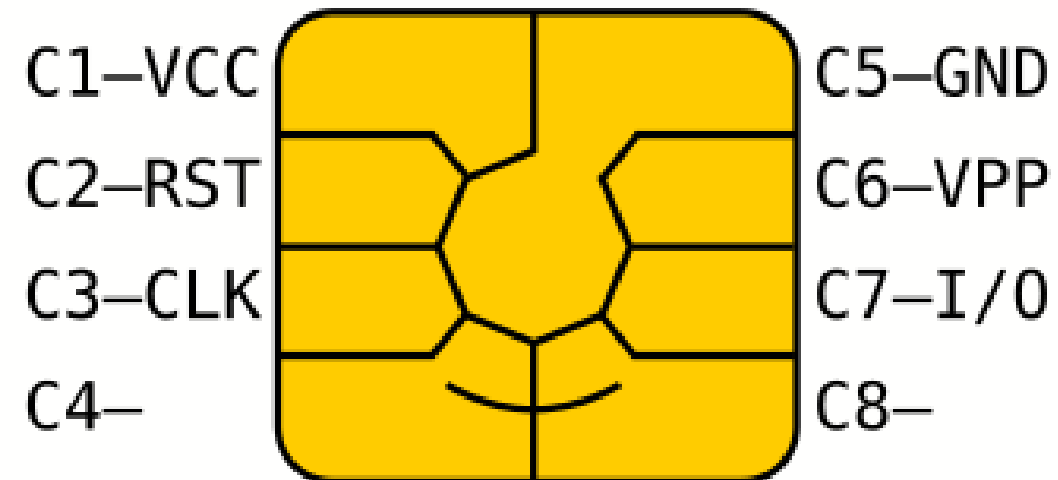
- CVV Types

| Type                                                   | Description                                                                                                     | Use Case                          | Service Code      |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------|-------------------|
| CVC1 / CVV1                                            | 3-digit code encoded on track 1 and 2 of the magnetic stripe. Used in card-present transactions with signature. | Card-present (swipe) transactions | Encoded on stripe |
| CVV2 / CVC2                                            | 3-digit code printed on the card. Used for card-not-present transactions like online purchases.                 | Online and remote transactions    | 000               |
| iCVV                                                   | Code generated electronically by contactless and/or EMV chip cards.                                             | Contactless and chip transactions | 999               |
| CDCVM (Consumer Device Cardholder Verification Method) | Verification using biometrics or device passcode (e.g., Apple Pay, Google Pay).                                 | Mobile-based payments             | -                 |



# Elements in Transaction Cards

- **EMV Chip**



Contact pad for the electrical interface on the front side of a credit card

**C1: VCC:** Power supply.

**C2: RST:** Reset signal, used to reset the card's communications.

**C3: CLK:** Provides the card with a clock signal, from which data communications timing is derived.

**C5: GND:** Ground (reference voltage).

**C6: VPP:** ISO/IEC 7816-3:1997 standard designated this as a programming voltage: an input for a higher voltage to program persistent memory.

**C7: I/O:** Serial input and output (half-duplex).

**C4, C8:** The two remaining contacts are AUX1 and AUX2 respectively and are used for USB interfaces and other uses.



# Elements in Transaction Cards

- **Magnetic Stripe**

A magnetic stripe card stores data on magnetic material attached to a plastic card. It is read by swiping through a magnetic reader and is used for credit cards, ID cards, and transport tickets. Some cards may also have RFID tags, transponders, or microchips for access control or electronic payment.

## Tracks on Magnetic Stripe Cards:

- **Track 1:** Higher bit density (210 bits/inch), includes the cardholder's name and alphabetic text. Written using DEC SIXBIT code with odd parity.
- **Track 2:** Lower bit density (75 bits/inch), contains essential transaction data.
- **Track 3:** Rarely used; may not be present on all cards.

Financial card data on Track 1 follows specific formats: A (proprietary use), B (common format), C-M (ANSI use), and N-Z (issuer-specific).



Front side of the first Magnetic Stripe plastic credit card.

# Elements in Transaction Cards

## • Magnetic Stripe

### Track 1 (Format B) Structure:

- Start sentinel: % (1 character)
- Format code: "B" (1 alpha character)
- Primary account number (PAN): Up to 19 characters
- Field separator: ^ (1 character)
- Name: 2 to 26 characters (Surname separator: /)
- Field separator: ^ (1 character)
- Expiration date: YYMM (4 characters)
- Service code: 3 characters
- Discretionary data: PVKI (1 character), PVV (4 characters), CVV/CVC (3 characters)
- End sentinel: ? (1 character)
- LRC: Validity character calculated from track data

### Track 2 Structure:

- Start sentinel: ; (1 character)
- PAN: Up to 19 characters
- Separator: = (1 character)
- Expiration date: YYMM (4 characters)
- Service code: 3 digits (interchange rules, processing, services)
- Discretionary data: Same as Track 1
- End sentinel: ? (1 character)
- LRC: Validity character for input verification

# Elements in Transaction Cards

- **Service code values common in financial cards:**

## First digit

- 1: International interchange OK
- 2: International interchange, use [IC \(chip\)](#) where feasible
- 5: National interchange only except under bilateral agreement
- 6: National interchange only except under bilateral agreement, use IC (chip) where feasible
- 7: No interchange except under bilateral agreement (closed loop)
- 9: Test

## Second digit

- 0: Normal
- 2: Contact issuer via online means
- 4: Contact issuer via online means except under bilateral agreement

## Third digit

- 0: No restrictions, PIN required
- 1: No restrictions
- 2: Goods and services only (no cash)
- 3: ATM only, PIN required
- 4: Cash only
- 5: Goods and services only (no cash), PIN required
- 6: No restrictions, use PIN where feasible
- 7: Goods and services only (no cash), use PIN where feasible

# Elements in Transaction Cards

- **Programming**

Parsing Track 1 and Track 2 can be done with Regular Expressions.

Track 1:

```
^%B([0-9]{1,19})\[^\^]{2,26}\^([0-9]{4}|\^)([0-9]{3}|\^)([^\?]*)\?$
```

This Regex will capture all of the important fields into the following groups.

- Group 1: Payment card number (PAN)
- Group 2: Name (NM)
- Group 3: Expiration Date (ED)
- Group 4: Service Code (SC)
- Group 5: Discretionary data (DD)

Track 2

```
^\;([0-9]{1,19})\[^\^]{2,26}\^([0-9]{4}|\^)([0-9]{3}|\^)([^\?]*)\?$
```

- Group 1: Primary Account Number (PAN)
- =
- Group 2: Expiration date (ED) or =
- Group 3: Service code (SC) or =
- Group 4: Discretionary data (DD) or empty

# Elements in Transaction Cards

- **Contactless Technology (NFC).**

- Contactless payment is a secure, touch-free method using Near-Field Communication (NFC) and Radio-Frequency Identification (RFID) for quick transactions without physical contact.
- It eliminates the need for handing over a card, entering a PIN, or signing, making it fast and convenient.
- The exchange happens within 4 cm to prevent accidental readings, indicated by the Wi-Fi-like symbol on POS terminals.

- **How It Works:**

- Initialization
- Communication
- Encryption
- Authorization
- Confirmation



**EMV contactless symbol on compatible payment terminals (EMV: Europay, Mastercard, and Visa)**



# Elements in Transaction Cards

- **Advantages of Contactless Payment**

- Enhanced Customer Experience
- Better Operational Efficiency:
- No Additional Cost
- Fraud Protection

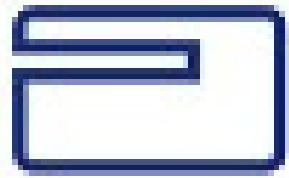
- **Challenges of Contactless Payment**

- Initial Setup Expenses
- Consumer Reluctance
- Limited Transaction Amounts



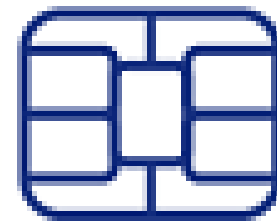
# Entry Modes

- Not all entry methods are available on all cards.



## Mag Strip

Black stripe on the back of most cards; swiping may prompt for chip insertion if an EMV chip is present.



## EMV

Secure chip on the front of the card (round or square) used for inserted transactions.



## Contactless

"Tap-to-pay" using NFC; payment transfers when the card is held near a compatible terminal. Also works with mobile and wearable devices.

- **Debit Transactions:** Must be swiped, tapped, or inserted — manual entry is not allowed.



# Types of Banking Accounts

1

## Current Account

A current account is designed primarily for business owners, traders, and entrepreneurs who need to make frequent transactions.

### Key Features:

- Overdraft Facility
- Free Transactions
- No Interest Earned
- Minimum Balance Requirement
- Ideal For: Businesses and professionals

# Types of Banking Accounts

2

## Savings Account

A savings account is a standard deposit account where you can save money while earning interest on your deposits.

### Key Features:

- Interest Earnings
- Limited Transactions
- Types of Savings Accounts: Regular Savings Accounts, Children's Savings Accounts, Senior Citizens' Savings Accounts, Women's Savings Accounts, Institutional and Family Savings Accounts
- Additional Features: Zero-balance accounts, auto sweep facilities, debit cards, bill payments, and cross-product benefits like discounts on demat accounts.
- Ideal For: Individuals looking to save money and earn interest

# Types of Banking Accounts

3

## Salary Account

A salary account is a type of savings account opened by employers for their employees to facilitate the direct credit of salaries.

### **Key Features:**

- No Minimum Balance
- Additional Benefits: higher transaction limits, special offers, lower loan interest rates.
- Reimbursement Accounts: Separate accounts for allowances and reimbursements.
- Ideal For: Employees who want a hassle-free way to receive their salary.

# Types of Banking Accounts

4

## Fixed Deposit Account

A fixed deposit (FD) account allows you to earn a higher interest rate on a lump sum amount deposited for a fixed period.

### Key Features:

- Fixed Interest Rate
- Premature Withdrawal: Generally not allowed without a penalty with some exceptions
- Safe Investment: Secure way to earn a guaranteed return.
- Ideal For: Individuals looking to invest a lump sum for a guaranteed return over a fixed period.

# Types of Banking Accounts

5

## Recurring Deposit Account

A recurring deposit (RD) account allows you to invest a fixed amount regularly and earn interest on it.

### Key Features:

- Regular Investments: monthly or quarterly.
- Fixed Tenure: Premature withdrawals attract a penalty.
- Interest Earnings: Interest is typically compounded quarterly.
- Ideal For: Individuals who want to save regularly and earn interest over a fixed period without investing a lump sum upfront.



# Types of Banking Accounts

6

## DEMAT Account

- Holds shares, bonds, and other securities digitally.
- Eliminates the need for physical certificates, making transactions quicker and safer.
- Facilitates easy buying, selling, and transfer of securities.
- Overseen by NSDL and CDSL in India.
- Reduces the risk of theft and loss by storing securities electronically.

# Types of Banking Accounts

7

## NRI Accounts

NRI accounts are designed for Indians living abroad, offering various options to manage their earnings and savings in India.

### Types of NRI Accounts:

A

#### Non-Resident Ordinary

- Rupee Account
- Taxable Interest
- Convertibility

B

#### Non-Resident External

- Repatriable Funds
- Tax-Free Interest
- Foreign Earnings Only

C

#### Foreign Currency Non-Resident

- Foreign Currency
- Tax-Free Interest
- Repatriable Funds

**Ideal For:** Non-resident Indians



# Types of Banking Accounts ( Based on Ownership)

1

## Individual Accounts

- Held by a single person.
- Allows deposits, withdrawals, and interest earnings.

2

## Joint Accounts

- Shared by two or more individuals.
- Equal access and ownership of funds.
- Commonly used by couples, families, or business partners.

3

## Business/Corporate Accounts

- Used by companies to manage finances.
- Also known as a business or company account.



***Thank  
you***

**- Regards**

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